APPLIED MATHEMATICS

AMATH 200s

AMATH 231 LEC,TST,TUT 0.50

Course ID: 003316

Calculus 4

Vector integral calculus-line integrals, surface integrals and vector fields, Green's theorem, the Divergence theorem, and Stokes' theorem. Applications include conservation laws, fluid flow and electromagnetic fields. An introduction to Fourier analysis. Fourier series and the Fourier transform. Parseval's formula. Frequency analysis of signals. Discrete and continuous spectra. [Offered: F,W,S]

Prereq: MATH 237 or 247.

Antireq: MATH 207, 212/ECE 206, MATH 217, 227

AMATH 242 LAB,LEC 0.50

Course ID: 011363

Introduction to Computational Mathematics

A rigorous introduction to the field of computational mathematics. The focus is on the interplay between continuous models and their solution via discrete processes. Topics include pitfalls in computation, solution of linear systems, interpolation, discrete Fourier transforms, and numerical integration. Applications are used as motivation.

[Note: This course may be substituted for CS 370 in any degree plan or for prerequisite purposes; lab is not scheduled and students are expected to find time in open hours to complete their work. Offered: W,S]

Prereq: (One of CS 116, 136, 138, 146), MATH 235 or 245, 237 or 247.

Antireq: CS 335, 370, ECE 204, MTE 204

(Cross-listed with CS 371)

AMATH 250 LEC,TST,TUT 0.50

Course ID: 003317

Introduction to Differential Equations

Physical systems which lead to differential equations (examples include mechanical vibrations, population dynamics, and mixing processes). Dimensional analysis and dimensionless variables. Solving linear differential equations: first- and second-order scalar equations and first-order vector equations. Laplace transform methods of solving differential equations. [Offered: F,W,S]

Prereq: (One of MATH 106, 114, 115, 136, 146, NE 112) and (One of MATH 118, 119, 128, 138, 148).

Antireq: AMATH 251, 350, MATH 218, 228

AMATH 251 LEC,TUT 0.50

Course ID: 014120

Introduction to Differential Equations (Advanced Level)

AMATH 251 is an advanced-level version of AMATH 250. This course offers a more theoretical treatment of differential equations and solution methods. In addition, emphasis will be placed on computational analysis of differential equations and on applications in science and engineering.

[Note: AMATH 251 may be substituted for AMATH 250 whenever the latter is a requirement in an Honours plan. Offered: F]

Prereq: (One of MATH 106, 114, 115, 136, 146, NE 112) and (One of MATH 118, 119, 128, 138, 148).

Antireq: AMATH 250, 350, MATH 218, 228

AMATH 271 LEC 0.50

Course ID: 013864

Introduction to Theoretical Mechanics

Newtonian dynamics, gravity and the two-body problem, introduction to Lagrangian mechanics, introduction to Hamiltonian mechanics, non-conservative forces, oscillations, introduction to special relativity. [Offered: F]

Prereq: (One of MATH 128, 138, 148), PHYS 121.

Coreq: (One of AMATH 250, 251, MATH 228), (One of MATH 227, 237, 247).

Antireq: PHYS 263

AMATH 300s

AMATH 331 LEC 0.50

Course ID: 003323

Applied Real Analysis

Topology of Euclidean spaces, continuity, norms, completeness. Contraction mapping principle. Fourier series. Various applications, for example, to ordinary differential equations, optimization and numerical approximation.

[Note: PMATH 351 may be substituted for AMATH/PMATH 331 whenever the latter is a requirement in an Honours plan. Offered: F,W]

Prereq: MATH 237 or 247

(Cross-listed with PMATH 331)

AMATH 332 LEC,TST,TUT 0.50

Course ID: 003324

Applied Complex Analysis

Complex numbers, Cauchy-Riemann equations, analytic functions, conformal maps and applications to the solution of Laplace's equation, contour integrals, Cauchy integral formula, Taylor and Laurent expansions, residue calculus and applications.

[Note: PMATH 352 may be substituted for AMATH/PMATH 332 whenever the latter is a requirement in an Honours plan. Offered: W,S]

Prereq: MATH 237 or 247.

Antireq: PHYS 365

(Cross-listed with PMATH 332)

AMATH 342 LAB,LEC 0.50

Course ID: 011451

Computational Methods for Differential Equations

An introduction to numerical methods for ordinary and partial differential equations. Ordinary differential equations: multistep and Runge-Kutta methods; stability and convergence; systems and stiffness; boundary value problems. Partial differential equations: finite difference methods for elliptic, hyperbolic and parabolic equations; stability and convergence. The course focuses on introducing widely used methods and highlights applications in the natural sciences, the health sciences, engineering, and finance. [Offered: F,W]

Prereq: (AMATH 242/CS 371 or CS 370) and (One of AMATH 250, 251, 350 or MATH 218, 228)

AMATH 343 LEC 0.50

Course ID: 003328

Discrete Models in Applied Mathematics

Difference equations and discrete dynamical systems. Mathematical models are taken from ecology, biology, economics, and other fields. [Offered: F]

Prereq: MATH 128 or 138 or 148

AMATH 350 LEC,TST 0.50

Course ID: 012744

Differential Equations for Business and Economics

First order ordinary differential equations. Applications to continuous compounding and the dynamics of supply and demand. Higher order linear ordinary differential equations. Systems of linear ordinary differential equations. Introduction to linear partial differential equations. The Fourier Transform and the diffusion equation. Discussion of the Black-Scholes partial differential equations, and solutions thereof. [Offered: F,W]

Prereq:(One of MATH106, 136, 146), 237 or 247, STAT 230or240 &(one of AFM 272/ACTSC291,ACTSC371,372,ECON371,BUS393W);Lev at least 3A;Not open to GenMath stdts.

Antireq:AMATH250,251,351,353,CIVE222, ENVE223,MATH211/ECE205,MATH218,228,ME203,PHYS364,SYDE211

AMATH 351 LEC,TUT 0.50

Course ID: 003329

Ordinary Differential Equations 2

Second order linear differential equations with non-constant coefficients, Sturm comparison, oscillation and separation theorems, series solutions and special functions. Linear vector differential equations in Rn, an introduction to dynamical systems. Laplace transforms applied to linear vector differential equations, transfer functions, the convolution theorem. Perturbation methods for differential equations. Numerical methods for differential equations. Applications are discussed throughout. [Offered: F,S]

Prereq: AMATH 250 or 251 and MATH 237 or 247; Level at least 3A.

Antireq: AMATH 350

AMATH 353 LEC,TUT 0.50

Course ID: 003330

Partial Differential Equations 1

Second order linear partial differential equations - the diffusion equation, wave equation, and Laplace's equation. Methods of solution - separation of variables and eigenfunction expansions, the Fourier transform. Physical interpretation of solutions in terms of diffusion, waves and steady states. First order non-linear partial differential equations and the method of characteristics. Applications are emphasized throughout. [Offered: W,S]

Prereq: AMATH 231 and (one of AMATH 250 or 251, MATH 211/ECE 205, MATH 218, 228).

Antireq: AMATH 350, PHYS 364

AMATH 361 LEC 0.50

Course ID: 003331

Continuum Mechanics

Stress and strain tensors; analysis of stress and strain. Lagrangian and Eulerian methods for describing flow. Equations of continuity, motion and energy, constitutive equations. Navier-Stokes equation. Basic equations of elasticity. Various applications. [Offered: W]

Prereq: AMATH 231 and (AMATH 271 or PHYS 263).

Coreq: AMATH 351 and (AMATH 353 or PHYS 364)

AMATH 373 LEC 0.50

Course ID: 003338

Quantum Theory 1

Critical experiments and old quantum theory. Basic concepts of quantum mechanics: observables, wavefunctions, Hamiltonians, and the Schroedinger equation. Uncertainty, correspondence, and superposition principles. Simple applications to finite and extended one-dimensional systems, harmonic oscillator, rigid rotor, and hydrogen atom. [Offered: W]

Prereq: AMATH 231 and (AMATH 271 or PHYS 263) and PHYS 234.

Antireq: PHYS 334

AMATH 382 LEC,TUT 0.50

Course ID: 011910

Computational Modelling of Cellular Systems

An introduction to dynamic mathematical modeling of cellular processes. The emphasis is on using computational tools to investigate differential equation-based models. A variety of cellular phenomena are discussed, including ion pumps, membrane potentials, intercellular communication, genetic networks, regulation of metabolic pathways, and signal transduction.

[Note: Offered in the winter of even numbered years.]

Prereq: One of MATH 118, 119, 128, 138, 148; Third year standing in an Honours plan

(Cross-listed with BIOL 382)

AMATH 383 LEC 0.50

Course ID: 014749

Introduction to Mathematical Biology

An introduction to the mathematical modelling of biological processes, with emphasis on population biology. Topics include ecology, epidemiology, microbiology, and physiology. Techniques include difference equations, ordinary differential equations, partial differential equations, stability analysis, phase plane analysis, travelling wave solutions, mathematical software. Includes collaborative projects and computer labs.

[Note: Offered in the winter of odd numbered years.]

Prereq: (One of MATH 106, 114, 115, 136, 146) and (One of AMATH 250, 251, 350 or MATH 218, 228) and (One of STAT 202, 206, 211, 220, 230, 231, 241)

AMATH 390 LEC 0.50

Course ID: 014564

Mathematics and Music

An introduction to some of the deep connections between mathematics and music. Topics covered include acoustics, including pitch and harmonics, basic Fourier analysis, the mathematics behind the differing pitch and timbre of string, wind and percussion instruments, scales and temperaments, digital music, musical synthesis.

[Note: Offered in the fall of even years.]

Prereq: One of MATH 118, 119, 128, 138, 148; Level at least 3A

AMATH 391 LEC 0.50

Course ID: 012282

From Fourier to Wavelets

An introduction to contemporary mathematical concepts in signal analysis. Fourier series and Fourier transforms (FFT), the classical sampling theorem and the time-frequency uncertainty principle. Wavelets and multiresolution analysis. Applications include oversampling, denoising of audio, data compression, and singularity detection.

[Note: Offered in the fall of odd years.]

Prereq: (One of AMATH 231, ECE 342, PHYS 364, SYDE 252) and (One of MATH 114, 115, 136, 146, SYDE 114)

AMATH 400s

AMATH 442 LAB,LEC 0.50

Course ID: 011448

Computational Methods for Partial Differential Equations

This course studies several classes of methods for the numerical solution of partial differential equations in multiple dimensions on structured and unstructured grids. Finite volume methods for hyperbolic conservation laws: linear and nonlinear hyperbolic systems; stability; numerical conservation. Finite element methods for elliptic and parabolic equations: weak forms; existence of solutions; optimal convergence; higher-order methods. Examples from fluid and solid mechanics. Additional topics as time permits. [Offered: F]

Prereq: AMATH 342

AMATH 451 LEC 0.50

Course ID: 003354

Introduction to Dynamical Systems

A unified view of linear and nonlinear systems of ordinary differential equations in Rn. Flow operators and their classification: contractions, expansions, hyperbolic flows. Stable and unstable manifolds. Phase-space analysis. Nonlinear systems, stability of equilibria and Lyapunov functions. The special case of flows in the plane, Poincare-Bendixson theorem and limit cycles. Applications to physical problems will be a motivating influence. [Offered: W]

Prereq: AMATH 351

AMATH 453 LEC,TUT 0.50

Course ID: 003355

Partial Differential Equations 2

A thorough discussion of the class of second order linear partial differential equations with constant coefficients, in two independent variables. Laplace's equation, the wave equation and the heat equation in higher dimensions. Theoretical/qualitative aspects: well-posed problems, maximum principles for elliptic and parabolic equations, continuous dependence results, uniqueness results (including consideration of unbounded domains), domain of dependence for hyperbolic equations. Solution procedures: elliptic equations -- Green functions, conformal mapping; hyperbolic equations -- generalized d'Alembert solution, spherical means, method of descent; transform methods -- Fourier, multiple Fourier, Laplace, Hankel (for all three types of partial differential equations); Duhamel's method for inhomogeneous hyperbolic and parabolic equations.

[Note: Offered in the fall of odd years.]

Prereq: AMATH 351 and 353

AMATH 455 LEC 0.50

Course ID: 003356

Control Theory

Feedback control with applications. System theory in both time and frequency domain, state-space computations, stability, system uncertainty, loopshaping, linear quadratic regulators, and estimation. [Offered: W]

Prereq: (AMATH/PMATH 332 or PMATH 352) and AMATH 351

AMATH 456 LEC 0.50

Course ID: 003357

Calculus of Variations

Concept of functional and its variations. The solution of problems using variational methods - the Euler-Lagrange equations. Applications include an introduction to Hamilton's principle and optimal control. [Offered: F]

Prereq: MATH 237 or 247 and (One of AMATH 250 or 251, MATH 211/ECE 205, MATH 218, 228); Level at least 3B

AMATH 463 LEC,TUT 0.50

Course ID: 003359

Fluid Mechanics

Incompressible, irrotational flow. Incompressible viscous flow. Introduction to wave motion and geophysical fluid mechanics. Elements of compressible flow. [Offered: F]

Prereq: (AMATH 353 or PHYS 364) and AMATH 361.

AMATH 473 LEC,TUT 0.50

Course ID: 003369

Quantum Theory 2

The Hilbert space of states, observables, and time evolution. Feynman path integral and Greens functions. Approximation methods. Co-ordinate transformations, angular momentum, and spin. The relation between symmetries and conservation laws. Density matrix, Ehrenfest theorem, and decoherence. Multiparticle quantum mechanics. Bell inequality and basics of quantum computing. [Offered: F]

Prereq: AMATH 373 or PHYS 334

(Cross-listed with PHYS 454)

AMATH 474 LEC,TUT 0.50

Course ID: 015892

Quantum Theory 3: Quantum Information and Foundations

Theory of correlations and entanglement; theory of quantum channels, detectors; the measurement problem, in quantum mechanics; phase space formulation of quantum mechanics; entanglement in infinite dimensional quantum systems; introduction to open quantum systems; and exploration of current research directions in quantum information. [Offered: W]

Prereq: AMATH 473/PHYS 454; Level at least 4A Mathematics or Science students only

(Cross-listed with PHYS 484)

AMATH 475 LEC,TUT 0.50

Course ID: 003371

Introduction to General Relativity

Tensor analysis. Curved space-time and the Einstein field equations. The Schwarzschild solution and applications. The Friedmann-Robertson-Walker cosmological models. [Offered: W]

Prereq: (AMATH 231 or MATH 227) and (AMATH 271 or PHYS 263); Level at least 4A Honours Mathematics or Science students

(Cross-listed with PHYS 476)

AMATH 477 LEC,TUT 0.50

Course ID: 015788

Stochastic Processes for Applied Mathematics

Random variables, expectations, conditional probabilities, conditional expectations, convergence of a sequence of random variables, limit theorems, minimum mean square error estimation, the orthogonality principle, random process, discrete-time and continuous-time Markov chains and applications, forward and backward equation, invariant distribution, Gaussian process and Brownian motion, expectation maximization algorithm, linear discrete stochastic equations, linear innovation sequences, Kalman filter, various applications.

[Note: Offered in the fall of odd numbered years.]

Prereq: (One of AMATH 250, 251, 350, MATH 211/ECE 205, MATH 218, 228), STAT 230 or 240

AMATH 495 LEC 0.50

Course ID: 003382

Reading Course

Reading course as announced by the department.

Prereq: Not open to General Mathematics students